

hi everyone in this video i'll be trying  
to make sense of  
what these terms ontology and  
epistemology  
actually mean what question and  
challenge are they addressing  
so that's the learning objective just  
really understanding what do these  
concepts  
and questions even mean and google text  
continuously uses these so it's  
important  
for be able to make proper sense of  
**Guba** it's important to  
be sure that we have a good  
understanding of what challenge are we  
actually  
thinking about when we're thinking about  
epistemology and ontology  
and so briefly put it's  
what world are we theorizing  
about so what is it  
what's the status of this thing that  
we're looking out at  
it's ontology and epistemology is then  
something about how can we access this  
this wold , how do we actually know

anything if the statement means  
knowledge in old greek and ancho means  
being  
so what's the status of being one o  
of these social science concepts and  
epistemy means  
the knowledge what do we actually know  
and one could argue that in  
Uncovering Reality?  
physics it's actually possible to really  
uncover  
reality one could make the argument that  
it's like a treasure hunt  
and we're actually able to uncover the  
mathematical and 100 correct laws  
prescribing the physical behavior of  
atoms and planets etcetera one could in  
principle try to make that argument  
because it  
almost seems like we are uncovering  
uh these laws and as ricky gervais said  
if we burn all physics books and then  
sort of  
start over again we should actually more  
or less  
end up in reasonably similar  
descriptions this equation  
should be the same and our understanding  
of atoms should be the same so one could  
sort of argue that we are uncovering  
what reality actually is

when we then get to whether it's  
motivation or  
um any kind of equations we might work  
with in the social science about how  
values of something evolves it's not  
as clear that we can say well now we  
uncover the reality of motivation  
we still have a bunch of different ways  
of trying to measure this and make sense  
of this and  
predict how motivated our people are um

so it doesn't really in the same way  
seem to be the case that we can call  
this a  
treasure hunt of uncovering the actual  
law and finding it  
in the box somewhere um  
it it seems to be the case that there

are at least some elements of shows of  
science that is different  
that's really what the issue is about  
what what is  
this thing we call motivation what what  
status of being does it have  
and how can we access it um  
and then again the point is that it's  
somewhat easier  
to uncover and  
claim that we're uncovering how the

world really looks like in physics  
because in the social sciences we have  
to decide how we define  
unemployment and we have to make some  
assumptions and decisions and  
two unemployed people can be different  
in many many different ways  
two hydrogen items are not going to be  
different in many ways  
um but let me sort of try to this is  
sort of what the  
contents mean and what questions they uh  
represent but let me try to get a little  
bit more into what challenge we're  
facing here  
and this is just to link back to this is  
a very very old discussion for anyone  
who studied some kind of philosophy and  
Plato  
Cave Allegory it's basically the same  
thing what is the status of truth  
beauty and justice those were the terms  
that played you were interested in  
we are in social science of course  
interested in different kinds of  
concepts but it's still in principle the  
same thing what what  
what is what are these things what what  
how do they exist do they exist out  
there  
"Yes, Prime Minister" clip

let's try to illustrate this by  
watching something a bit more  
contemporary than Plato even though that  
it's still a couple of decades old so  
you might not know this series it was  
very popular  
um a couple of decades ago let's watch  
these two people discuss something for a  
couple of minutes it's from the show „yes  
prime minister”  
they're discussing national  
service  
in in Danish  
so he's going to say something new and  
”adical in the broadcast but that silly  
grand design  
bernard that was precisely what you had  
to avoid  
how did this come about i shall need a  
very good explanation well he's very  
keen on it  
what's that got to do with it things  
don't happen just because prime  
ministers are very keen on them  
neville chamberlain was very keen on  
peace  
who thinks he thinks it's a vote winner  
ah  
that's more serious dan what makes him  
think that  
well the party have had an opinion

pogden it seems all the voters are in  
favor of bringing back national service  
well i have another opinion pearl done  
showing the voters are against bringing  
back national services  
we can't be foreign have you ever been  
surveyed  
yes well not me actually my house oh i  
see what you  
know what happens nice young lady comes  
up to you obviously you want to create a  
good impression you don't want to look  
at fool do you  
no so she starts asking you some  
questions  
mr willie are you worried about the  
number of young people without jobs  
yes are you worried about the rise in  
crime among teenagers  
yes do you think there's a lack of  
discipline in our comprehensive schools  
yes do you think young people welcome  
some authority and leadership in their  
lives  
yes do you think they respond to a  
challenge yes  
would you be in favor of reintroducing  
national services  
oh well i suppose i might yes or no  
yes of course you would benefit after  
all you've told you can't say no to that

so they don't mention the first five  
questions they publish the last one  
he said really what they do well not the  
reputable ones no but there aren't many  
of those

alternatively the young lady can get the  
opposite result how

mr woolley are you worried about the  
danger of war yes

are you worried about the growth of  
armaments yes do you think there's a  
danger in giving young people guns and  
teaching them how to kill

yes do you think it's wrong to force  
people to take up arms against their  
will

yes would you oppose the reintroduction  
of national surveys

yes”

Two Different Challenges of Independent Opinion

and the reason i'm then drawing upon  
such a video as this

and in this class is it could make us  
think about

does an independent opinion on national  
service

actually exists is there

something out there that a proper survey

a proper survey questions could get to

or is the challenge that we simply don't

i mean maybe people actually have an

opinion

but it's just really really difficult to  
get to so we have to accept that our  
response  
is always going to be sort of imperfect  
so that's

sort of two different challenges one  
could argue that these are two different  
challenges is it

because there is nothing in our i  
don't know in our brains that we can try  
to

to dive into or is it because even if  
there was something in our brain that  
signified whether we have enough what  
opinion or national service we have we  
just simply can't get to it because  
whatever way we ask questions  
we risk sort of shaping the answer we  
get and maybe getting another answer  
than was intended

um this is and also the way that it's  
being described by this

tweeter it's a sort of very famous  
pollsters and

a quite famous pollster in american context  
um

and and he's saying a pollster's 's job should  
be to find out about already existing  
opinions not create ones that already  
exist



and then the point that is being made  
here in this video is to say well  
it's actually quite difficult to claim  
that there is something out there  
um that we're trying to get to because  
no matter how we phrase our question we  
risk sort of shaping whatever  
thought people have so the argument here  
is that in contrast to when we saw the  
video with the light poles where we can  
say there is a light pulse out there and  
we are just trying to capture it as  
adequately as possible  
like the treasure hunt example as if  
there is something out there that we can  
get to  
the point with this prime minister or a  
national service example is to say  
it's not clear at least that there is  
something independent out there  
already existing that we're just trying  
to get to because  
no matter how we shape and frame our  
questions we risk influencing the  
answers we get  
as is being said in the video of course  
we can try to do a decent and good  
and transparent job about it not hiding  
the first five questions  
but that doesn't change the fact that  
it's simply a different kind of thing

as in in many areas in the physical  
sciences

Measurements and Measured Object

and i'm going to use an actually a  
danish example of Tycho Brahe to sort of  
try to further

illustrate that um Tycho Brahe  
was developing sort of a place to  
look out into the universe from his  
his island here in denmark and  
at some point he realized that his  
observatory

shook slightly in the wind making his  
measurements imperfect  
so he could move his instruments into  
the underground bunkers and then he sort  
of continues to optimize that i think  
that's really interesting

example to draw upon here because here  
it was quite clear from him he could  
sort of clearly identify that wind  
influences his measurements so if you  
went to an underground bunker there was  
less wind and

sort of less external interference and  
if we then

try to make the same thing and when we  
measure attitudes about national service  
or unemployment or stock markets etc  
it's just not clear that we in the same  
way can say ah let's just move the way

we ask questions

to this angle and then we've solved all

our problems

so this is to illustrate that to go

could

sort of clearly optimize the measurement

of planets the planets are out there

it's the way he's trying to measure it doesn't influence where the planet is

planet doesn't care about the position

of his

telescope maybe

the one's motivation or attitude and all

these things

it's not as clear that they're

completely independent from the way that

we're trying to measure it

as this was nelson's point as well so

the planet

the measurement of a planet is not i

mean the planet's

position in the universe is not

influenced by our telescope

um and that's sort of a different

situation that we're in and for Brahe

in other words it's easier to claim this

ontological distinction

between measurement and what is measured

so it's more clear that two

these are two distinct things while

the difference between the unemployed

person and the way we define

unemployment i mean these two things are

intricately are clearly interrelated

um and this is again the particular

challenge we're facing in the business

administration

then other fields are facing differently

i should although

add that i've mainly talked about

classical physics and maybe some of you

have already heard about quantum physics

being sort of different

it's actually it's it's actually the

case in quantum physics that we can't

both measure the position

of a small little thing like a quant

and its speed its momentum we simply

cannot know both things at once because

the moment we try to measure the

position

we are influencing the momentum when the

speed that is

this little quant has so actually there

is a kind of an

isological limit in in quantum physics

um

where it's simply not possible to to

make these precise measurements because

no matter how you try to measure it

you're influencing whatever you measure

same thing no matter how you try to

define innovation or firm performance or

unemployment

you're making some assumption the

decisions that influences whatever

result you're getting out of it

## Summary of Ontological and Epistemological Issues

so that's the ontology and epistemology

issues

uh these very abstract issues that i've

tried to deal with here

is for one what's the status of the

social science concepts and theories

and i hope i've sort of been able to

show that

Tycho Brahe

can sort of claim that he's trying to

measure something that is independent

from his telescope

it's off more difficult in our business

administration

administration field so it's it's more

difficult to say that we sort of

are measuring something independent and

real and separate from us

and then furthermore epistemology it's

it's it's not

clear that we can sort of easily access

this without interfering so

Tycho Brahe

had the challenge as well

that he had to take account of the wind

and maybe he has to take

other things into account i don't know

when the earth is moving a little bit in

his underground bunker

so they're still facing the challenge

but it might be a bit easier to make



sure that

we are measuring exactly what we're

going for rather than the national

service example because

how do we and then know if we've

actually measured people's opinion on

national service or how motivated they

are

or whether we've actually measured

unemployment or firm performance in the

end it's

us researchers that decide whether

something is the best

most true measurement

so this is why i've used this contrast

to physics because i think intuitively

it's more easy to understand

than an atom or a planet it's something

independent from us and it's out there

and we can get to it and we can

uncover a theory of gravity and very

likely that if we burn all our physics

books and then recreate them a thousand

years from now we'll get to the same

answers

it's it's less clear that this is going

to be the case in

study of unemployment supply demand

efficient market hypothesis, motivation or

whatever

um business administration concept that

we can think about it's not as clear

that we have these distinctions this was

also what nelson was talking about

now we're just getting sort of a

different kind of terminology

that also helps us understand what

Guba's book chapter is about

..ñ<sup>3/4</sup>→□NΠΠΠΠم☺☺☺♣→Π<sup>3/4</sup>ð-ø☺İ=ÞÑñ□=Þ  
♣x<sup>3/4</sup>□xΠΠ □İ| 5ðÑττ<sup>3/4</sup>9İΠ=Π<sup>3/4</sup>7<sup>3/4</sup>þ<sup>3/4</sup>Ñ-Π%  
%

hi everyone and welcome to lesson six

today we'll be reflecting on what a

theory is at a very abstract level

really characterizes today's lesson is

that it's not

just sort of difficult to understand

what the answer is but even

understanding the question

and the issue at hand can be can be

tricky

so just so you know already in advance

you might not exactly know

and figure out what these questions even

mean from the beginning  
but during the lecture and and hopefully  
at the end of  
this and the next lecture it should  
become more clear what's going on  
so we'll be trying to answer these  
questions here how do we know something  
about the world how certain can we be  
about the social science world that we  
make theories about and what is this  
social science what after all  
we call these issues epistemology and  
ontology and we'll get more back to  
to those terms later on but here  
initially i want to  
use Nelson to make a contrast to physics  
and some natural sciences  
in order to make sense of what the deal  
in social science what situation  
are we in so we will  
be making that comparison and contrast  
to physics  
because it will tell us a lot about what  
situation that we are in here in the  
social science  
Measuring atoms vs. social science constructs  
and sort of very briefly and crudely put  
on an overall level  
it can be illustrated and highlighted  
via this example that i already  
showed in in lesson one we have a very

good idea of what an atom looks like  
this  
simple version of an atom and we  
have that theoretically but we also  
have that because we can we can  
empirically really measure  
what one atom looks like we can sort of  
take a picture we'll get a little bit  
more back to that in the Kuhn lesson in  
lesson nine but  
um we have a very good idea of what an  
atom looks like and whether we'll check  
this atom or that atom or another atom  
in in another part of the globe or  
another part of the galaxy or universe  
they will basically be identical and we  
will have a very  
good idea of it and we can point to it  
we say look here it is  
while let's consider the efficient  
market hypothesis or any  
theory on motivation we can't to the  
same clear degree say  
look this is where the essence of a  
motivation is we can't  
point to any place in the human body  
whether in  
consciousness or physically and say this  
is  
what motivation is it's just a different  
kind of

construct it's a different thing um  
and this is what we then will want to  
address and it's the same with the  
efficient market hypothesis  
we might have a good theoretical  
understanding sort of roughly what it is  
and we can use it to explain  
today's capitalistic society and how  
values were sort of emerge and unfold  
but  
we can't sort of point to somewhere in the  
Stockholm market and say  
look here it is and sometimes the market  
appears to be less efficient than they  
should be so  
what's the deal with that um atoms they  
they behave  
always the way that we expect them to do  
there is something particular about the  
social sciences  
that makes it different to sort of this  
atom comparison and this is what the  
Unlike Physics...  
nelson text is about  
he's trying to make us reflect on what  
kind of science is  
economics or business administration and  
and one of the key things he wants to  
say is that we  
sometimes hear whether it's politicians  
or scientists to just delay

understanding

saying well physics that's the proper science and then if something can't be sort of as robust and clear as what we find out in physics well then

it's not really a proper science in contrast Nelson is saying we should not model social science on physics we and he's explaining why the social sciences are different and why these things are important

and he's also pointing out there are actually a range of natural sciences that aren't like physics so

if we sort of have the classical physics expectations of being able to predict as we can

in classical physics well then biology and meteorology etc are not sort of i'm not cutting it as well they're not um

as clear as as a physics science so so this is what he's aiming for he wants us to think about how certain can we be about the conclusions we draw on business administration

and why is there sometimes uncertainty and why is that okay

so how can we talk about business administration as a science

and again the contract we're making is

and the question we could ask ourselves  
is why can't social science  
lead us to newtonian law like in physics  
we can use the theory of gravity to  
in this equation to very very clearly  
and accurately predict  
movement of any kind of a movement of  
objects or for instance the locations of  
planets  
um and the thing here with planets or  
atoms i mean we can  
describe them sort of completely in  
numbers as long as we know the weight  
and the um and the density and the for  
instance the speed of a planet  
and and then we measure the speed of all  
the other planets in a given solar  
system then we can exactly and precisely  
predict where jupiter is going to be  
230 000 years and four months and seven  
days and three hours and five seconds  
from now  
we we can do that to that extreme degree  
of precision  
um and and we can do that because we can  
as mentioned  
exactly measure what a planet is we can  
sort of define the essence of a planet  
while we can also ignore all the other  
stuff that is out there in the solar  
system there is galactic dust and there



are comments and in principle we could  
send out spaceships to try to interfere  
with these planetary movements but it's  
just i mean they're just too small and  
insignificant  
and then we can just ignore them um we  
can still make these very good  
predictions even though we ignore all  
these factors  
so this is just to say this is the  
comparison we in principle could make  
this is the state of the on in physics  
we have this homogeneous phenomena we  
explained and measured one atom hydrogen  
atom we measured them all  
and we can ignore a bunch of factors  
let's keep that in mind when we try to  
understand we try to understand the  
social sciences  
but before that also just sort of want  
to highlight the point that nelson is  
also making  
there are a bunch of natural sciences  
that aren't like physics  
any storm is not like any other storm i  
mean they're all  
different in some significant way it's  
not that we can't sort of try to say  
this the storm is here and we can make  
reasonable good predictions of where  
it's heading

as this link is talking about but  
it's not like one hydrogen atom is  
identical to another one  
it's not like one storm is identical to  
the other one um  
and there are also a bunch of different  
things that could potentially influence  
where a storm is moving i mean  
meteorology is still a pretty precise  
science it it still does rather well  
um and it's no surprise that it's  
snowing today when i  
when i'm making this video because  
meteorologists had predicted that  
so just to say physics shouldn't be used  
as a prototype for all kinds of sciences  
because  
i mean even other natural sciences  
aren't as precise and  
and we can't sort of make the clear  
causal predictions as  
as we can in physics and many other  
natural sciences so this is then getting  
Heterogeneity, Numerous influences & Change  
to a key  
uh a key slide and a key three key  
elements  
of the nelson paper first of all what  
he's trying to say is that the separate  
matter that we're studying in social  
sciences

they sort of fall into different classes  
and there's heterogeneity in each class  
if we want to study human beings  
they are not all just completely  
identical they're going to be very  
different if we study companies they're  
different  
if we study entrepreneurs they're  
different if we  
we we we can't sort of  
reduce them to a very small number of  
well-defined variables as we can with a  
planet and just say  
give me these two three pieces of  
information about a planet and i can  
make all the predictions that you want  
it's simply impossible when we talk  
about firms and people and  
countries and stock markets etc um  
so we can't as easily define the concept  
that we're dealing with  
what is um an innovative company or what  
is a motivated human being or  
it's just it's not that we can't say  
anything about it but it's not like with  
hydrogen atom we could just completely  
draw it and have a perfect understanding  
of it  
so that's the the units that we're  
interested in but also  
the things that then sort of influence

and we we can then talk about well what influences

motivation or a company's performance

and then

Nelson is making the point there's a

bunch of different things that influence

stuff in the social sciences there are

numerous highly variable and often can't

be separated sharply from one another

um if we're interested in firm

performance we can't just sort of say

okay let's look at competitors and what

they're doing

we have to for instance look at the

competitors we have to look at the

individuals in the company we have to

look at technological development we

have to look at the

global sort of economy we have to

consider whether Covid 19 crisis is just

going to influence

and sort of disrupt everything

i'll give a few more examples of this in

in just a bit but this is just sort of

the second

overall point that nelson is making for

one of the things we're interested in

uh heterogeneous um if we are comparing

a bunch of people and and

how well they're doing for instance in

the o'reilly experiment

they're not going to be completely identical the second point is in addition it's not as easy as with planetary movement just to say well there are only eight other planets and the sun that really influencing the way that the world revolves around the sun there is going to be a ton of things that we have to be interested in if we are thinking about firm performance and as this quote by famous physicist richard feynman says imagine how much harder physics would be if electrons had emotions i mean there's just a ton of different things that can influence human behavior and to make it even more complicated the stuff we study can even change over time we'll

get back to that in a later lesson

but that of course doesn't make it

easier that an atom is an atom and it

was the same thing

four billion years ago and will probably

be the same thing two billion years from

now

human beings companies and concepts etc

they can change

all the time so there is something about

the social science world that we're

studying that limits what we can do

and that's fundamentally different from

for instance the classical physics

let me give you a few more examples of

this so if we look at

biology and here there is someone who

really took

a bunch of e-coli cells and what is it

30 40 years ago

described in this wiki page he took

these 12 populations

of of these e coli and then sort of had

them in these different kinds of

vials and glasses and then he made them

sort of

evolve as they do they replicate

themselves and

and and as we can see over time this is

what we have on the x-axis we have the

number of generations

of e coli cells that have sort of been

reproducing over time

and on the y-axis we have the cell

volume so how many e coli cells were

there in each vial glass

and of course i mean not that they were

identical but it was pretty close the

number and cell volume

over 10 000 generations actually

continue to have been doing this over 50

000 generations um imagine that we let

12 populations of human beings develop

separately for 10 or 15 000 generations

i mean the variety would just be

absolutely enormous first of all many of

them

might have died out others will be

extremely big some will just barely have

survived

this is just not the reality we see in



biology and

it's not just something we can speculate

on someone has run the exact experiment

with the same and the same 12 original

population of e coli cells are still

evolving

in in his research lab and here is just

a quote that that

again makes the same point huge energy

of entrepreneurial phenomena

we have different kinds of people

starting economic activities for

different reasons different processes at

different paces different resources

different industries cultures economic

environments this is just the

heterogeneity of human in this case

entrepreneurial behavior we're looking

at

and this is a basic condition of social

science in general and business

administration in particular

and then if physicists had to deal with

that they could not come up with this

easy not easy but at least relatively

short and simple

theory of gravity so there's something

about the construct that we're

interested in

what does it mean to be unemployed an

example that nelson also uses

i mean that can mean you don't want a

job that can mean you have enough money

that you just focus on your

hobby it could mean that you're very

very actively searching for a job

it could mean you'll take a job if you

can find it it can mean that you're

working sorry not working but you're at

home taking care of your newborn kids

i mean being unemployed or seeking a job

can mean many many different things and

we might put

a bunch of people into that same

category but they will consist of

different motivations resources

interested and different willingness to

actually get a job

it's also using the innovation example

what does it mean for

a company to be innovative have they

sort of

developed something completely brand new

to the world or is it just a small

new addition and how do we even measure

whether something is innovative

um is it just based on how much many

products they can

sell or is it experts that have to

assess how innovative something is

it's just not clear that we can sort of

talk about the constructs as precisely

as we can in other fields

so this is also what he's saying in the

quote here at the underlying conception

of the phenomenon that we study they're

blurry edges they're constructed numbers

there isn't anything like in in physics

we might have struggled with

understanding

physics in the 17 18 19th century but

there was

something that we could get to when

unemployment or innovation

or all the other concepts we have in

business administration

it doesn't seem to be the case that

there's sort of one underlying

fundamental

thing that being unemployed is

so it's a constructed number um we make

assumptions when we make our

measurements

and we can sort of try to measure these

phenomena but it's a limited measure it

might be useful

but limited so we can make a definition

of what it means to be unemployed

and we can make a definition of

innovation and we can work with that

but it's useful but limited and as he

says this is something we need to be

aware of

in terms of how certain conclusions we

think we could come up with and what

kind of signs we

we can we can make

just to just to give you sort of a

simple example of this in

this is just the essence of what this is

doesn't really matter but this is a

typical

statistical analysis we have some

independent variables and we want to

predict

um some kind of dependent variable and

then we might say well whether people

have a university degree or not

is and might might have been an

important factor it doesn't seem to be

the case in this study

but it's not like sort of putting an

atom in as an independent variable and

then knowing that everyone in this

category are exactly the same while

having an university degree could be

from howard it could be from Aarhus, it

could be from

anywhere else in the world potentially

it i mean this would be a very

heterogeneous group that has been sort

of lumped into one overall

variable and it's not clear as nelson is

saying that there is



sort of one underlying clear conception

of something there that we're trying to

get to

we're constructing um our variable to

some degree we're deciding

is a bachelor degree university degree

or does it have to be a master's degree

do we include online universities we're

making assumptions we're making

decisions

as as we go along and that is the

essence of the kind of social science

that we have to do we have to make that

kind of um decisions

and on top of that i talked about how

the the

the entities that we study in social

science that can change over time

kieran healy makes a very succinct

point that we should think of social

science as a small bit like hurricane

forecasting

except the hurricane is able to watch

the news and change its behavior

accordingly

this is what we are facing in the social

science we can

have someone read theories and then

realize that that maybe i should change

behavior that's going to be more

efficient

and and then we can sort of respond to

whatever we are told and

to the theories that we have and we'll

get back to that in lesson 13

but um this is just already now to

indicate again another sort of

particular nature of

social sciences that's different from

meteorology for instance we're dealing

with self-reflecting human beings that

can decide to change behavior over time

and that just clearly doesn't make our

job any easier

Homo Economicus and (impossible) Simplifications

sort of the original economist

take on all this and you might have

heard this concept of Homo economicus

was actually quite literally to base

economics on physics as an idea if you

read some of the earliest

economic textbooks from the early 20th

century or late

19th century this is quite

literally what they did they were trying

to

copy and imitate physics and and so

there is this idea that we are dealing

with homo economicus and they are

rational and they only focus on money as

an incentive

and we're all basically like that maybe

emotions sort of come in the way but

that's just

because we're weak human beings indian

we are these rational human beings

and we can predict how we are going to

be influenced and that's the

dream that economics has had

and this was necessary to have that

that's also the point that if we want to

come up with

clear sort of mathematical uh clear

predictions of human behavior

then we need these homogeneous beings we

need to make these assumptions

um just as they've been able to do in

physics and we see here an example of it

here

and there's paper that i'm referring to

here so

there is paper the exact details are not

100 important

but there is this paper here that's

trying to find out if high leverage is

optimal for banks so should they

sort of spend the the

the money they have the kroner they have

in in many different types um

and and and and what they're then trying

to do in their model as you can see down

here is

that the model they're trying to create

the explanation they're trying to create

for establishing the higher bank

leverage is good they are ruling out

deposit insurance taxes and all other

distortionary factors

they are imposing these idealized

conditions i mean they're doing the

exact same

thing that and then sort of the whole

comparison and physics is also

illustrating that physics can do

when we're trying to predict planetary

emotions we can roll out all sorts of

stuff

galactic dust and comments etc because

they're not important

the question then is can we really rule

out

taxes agency deposit insurance etc when

we try to explain

whether banks should behave in this that

way that is not as obvious

as actually possible to make these

simplified assumptions

and and this is then again referring

back to nelson's three points

this is what he's trying to say that

there's a bunch of different things that

that influence whatever we're interested

in and it's just not easy to

isolate them and assume them away as



this paper has been trying to do

um because forces and conditions that

influence the subjects that we study

are numerous highly variable and often

cannot be separated sharply from one

another

this is what they've been trying to do

here nelson's point is

we can't easily do that we can't know

whether removing taxes from the equation

doesn't sort of disturb the whole

picture because they do influence

Limits of Experiments

another way of making this point is to

to reflect on what nelson has said about

experiments and the limits of

experiments

and and then we've talked about that the

the essence of the value of an

experiment is that

that different classrooms were probably

as good as identical

but nelson is here explicitly saying

that if we're sort of doing experiments

in educational setting

there are differences between schools

and there are even differences within

the school between classrooms

we can't be certain that

the entities we've been studying if we

for instance decide to do a randomized

controlled trial across

a bunch of schools and a bunch of

classrooms we're assuming that all these

different classrooms function in the

same way

but in practice they might not in the

elementary school example maybe some

classes opened the windows at 11 o'clock

and maybe some classes didn't open at 11

some classes maybe opened in at 12 and

maybe somewhere they ate lunch in a

different way than

the other place we we have to assume

that identical when we run our analysis

on the randomized control trial

but in practice there aren't really

nelson as you know again just saying

there's a bunch of things that influence

um

what we're interested in business

administration in a different way than

if we are studying planets or atoms

where we can make all these simplifying

assumptions so just to get back to this

Overview of Nelson (2016) and Reflection Exercise

slide that i've shown before this

overview really keep this in mind and

keep these three different elements in

mind

that this is what nelson is saying

subject matter that we study

it's heterogenous we can't just say

this is a good

company performance or this is what a

highly motivated person looks like

this is what an entrepreneur is they are

going to be different

there is a lot of heterogeneity and that

does create a challenge

furthermore not only are the concepts

that we're interested in

heterogeneous they're also being

influenced by a ton of different things

um it's not like an atom that can only

be influenced by very few forces in the

world

human beings and companies can be

influenced by

by competitors and global market

situations

and the health of the ceo or

the weather or whatever sort of we could

imagine there's just a ton of different

things that in the end could

influence what is going on it's not easy

to

separate them sharply from one another

um and again as mentioned the subject

matter changes

over time so there is a limit to what we

can do in social science and

simply because of the nature of the

world that we're interested in and we

need to sort of acknowledge that and

take that into account

when we try to figure out well what kind

of conclusions can we draw and get to in

the social sciences

here i've just added a bunch of quotes

that are relevant to sort of make sense

of what nelson is saying i won't go

through them in detail here um but but

i see them maybe just make this one

point that

we're trying to measure motivation and

leadership and relational culture and

strategies and supply and demand and

sometimes we can make fairly direct

measurements and sometimes we can't

um it sort of depends on the situation

and how easy it is to to measure what

we're interested in but we can never get

to the

precise measurement as what an atom is

or what a planet is

this is just for you to reflect um how

easy is it to measure whatever you're

interested in in your bachelor project

it's certainly not going to be as easy

as planets and

it's not going to be as easy to see what

influences whatever you're interested in

but

so it's worth thinking about what is it



that i'm actually interested in how heterogenous

are whatever variables i'm looking at

and how many different things can it

principle influence it

and does this thing does the stuff that

i'm interested in actually change over

time

that's the point that nelson is making

that we should reflect on and be aware

of

when we think about a given research

project so

Implications

this is sort of nelson's message here we

should not treat physics sort of

the ability to come up with natural laws

and perfect predictions as an idea

ideal because we are in a completely

different situation

we have very heterogeneous phenomena

we're interested in and

relations that we are trying to uncover

we just have to be very transparent and

honest about that

we shouldn't just say i measured

employment or unemployment we should

acknowledge well i measured unemployment

in this way and i realized that

this might be problematic because these

kind of unemployed people are not

included or

or i'm actually including unemployed

people that don't really want a job and

it doesn't really

count so being that's his point then

also one should be very transparent

about

the way that you're measuring whatever

you're interested in and acknowledge

whether it could have been measured in

different ways

and then may provide an argument for why

you did what you did

um and then also saying that

both quantitative and qualitative work

can be important it depends on what

you're going for

depends on the field that you're looking

at

and then there is a point that the more

homogeneous field is that there

sometimes it is possible to say well we

are only interested in

in money in this context anything else

is sort of irrelevant

maybe that is sometimes almost possible

but it's just not going to be as

obviously possible as in

in some natural sciences

and as this old quote from from

Aristotle published in 76 but of course written

thousands of years ago "we should not

expect more position in the treatment of

any subject than the nature of the

subject permits for

demanding logical demonstrations from a

teacher of rhetoric is clearly about as

reasonable as accepting mere plausability

from a mathematician”

so just sort of point is that we should

acknowledge what field social science

and business administration is

and then accept that we will

have reduced uncertainty because of

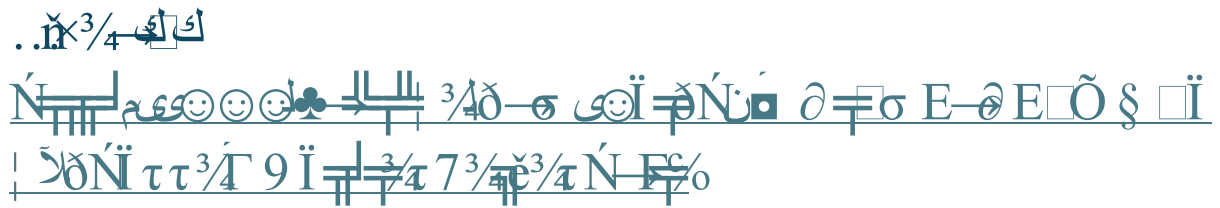
heterogeneity because of lots of

different things influencing

because stuff changes over time and that

is just the

situation that we we have to accept



hello a short note on the concepts of  
ontology and epistemology , I realize IS A  
tricky challenges to get your head  
around so I'll redo Guba's  
conceptualization of this what question  
is he asking and how you see answering  
them and I'll try to provide an  
additional metaphor an additional an  
example then hopefully can help you make  
make sense of it so first of all what is  
The question  
the question actually well for one it's  
the ontology what is this world  
then we investigating when we are  
looking at at atoms it's it's a little  
bit easier to think of it as something  
that is out there it's tangible we can  
measure it and it's even the same  
whether we are here in Denmark or  
anywhere else in the universe we will  
get the same measurement model and  
responses it makes sense to say or  
intuitively it's not implausible to say  
that these are real things that we

investigating and the positivist would  
then also say that's in the next step  
the epistemology would be to say well and  
we actually also have access to getting  
the right picture of what an atom looked  
like true knowledge objective knowledge  
should really correspond to the way the  
world really is there is something  
independent out there it's a little bit  
like the treasure hunt metaphor I gave  
that then that there is something out  
there that we can find and uncover and  
so theories not discovered and not  
invented or shaped or anything it's out  
there and then we have post-positivist  
who are slightly less optimistic they  
agree that yes there is something out  
there that we should try to make sense  
of not just in the sense that there are  
facts and stuff happening but also  
statical mechanisms via the  
supply-demand  
mechanism or how to motivate people that  
there is something out there to try to  
get you however this objectivity is is  
really just  
it is just an ideal we can't get there  
in the end  
it's it's gonna be approximations we can  
try to get as close to it as possible  
so we don't really find the treasure but

we'll what we're trying to get there  
that also says something about how we  
should think of the person doing the  
investigation and nor the investigator  
especially for a positivist we should of  
course be the fly on the wall it doesn't  
influence the data collection that's why  
we like an experimental approach it's  
trying to be as objective and distanced  
and neutral as possible post positivist  
but then say and we saw that I think to  
some degree in the Bernstein article  
that we can try to do as much as  
possible to make sure that the data we  
collect is not being shaped by how being  
shaped by how we ask questions so we  
hire research assistants that are going  
to help us collect data in a Chinese  
factory that are Chinese themselves so  
they can really blend in and the workers  
won't notice that something is going on if  
Bernstein done had been in there you  
would have clearly have influenced the  
potential responses because you would  
have been the odd one out  
so post-positivism would say yes let's  
have it as an idea let's ideal an idea  
let's try not to contaminate our data  
that's trying to be as objective as  
possible and we can still do qualitative  
data collection but it will should



always be about being as distance as  
neutral as possible as standardized as  
possible so a narrative approach for  
instance in your interview would be  
difficult in this framework because you  
want standardized responses that others  
would ideally also get if they asked the  
same question and then the  
constructivism angle that Guba is is  
is posing is then saying look you're  
saying just like the post positivist this  
week we can't step outside our and we  
can't step outside our worldview  
whatever we do to try to test if our  
theory of motivation  
job performance or whatever it is it's  
good it's it's taking departure in our  
history in our training in our education  
in our assumptions we can't step outside  
of this world so so we should actually  
just accept that then yes we want to  
investigate motivation and job  
performance we want to investigate how  
teacher at a university performs at his  
or her job we accept that there is no  
complete perfect and so to try to get to  
so we'll try to to to address different  
perspectives and try to get there in  
different way so of course the world  
exists and facts exist and stuff happens  
in the business world but the

supply-demand mechanism is not something independent independent of us in any sense it's something that has been created by our institutions our countries our business environment and and these could be different now I'll get back to this in a bit so we've

### Examples

already seen some of these examples here so this isn't really illustrate

what a post positivist would like to do the to cover our example was trying to get a sense of the solar system and comets the planets and in order to get a bit of measurement he went into the basement in order to not have the wind influence him and this wouldn't be the same ideal that people should have if they are interested in attitudes

unemployment stock markets innovation whatever we have in business administration field

try not to contaminate your data collection trying to standardize your thing a collection and then try to keep a distance then they yes prime minister video what this did was in to illustrate that this could actually really be difficult in some social science settings how can we be certain that first of all there actually is an answer

to in this case it was national service  
but it could have been anything else  
when we asked people how can we be  
certain it actually is an answer out  
there and and that's what sort of be the  
ontological angle of it is there  
something out there independent from our  
way of asking the epistemological angle  
is however we ask the question we risk  
influencing whatever kinds that we get  
as what sort of very crudely illustrated  
in the video  
that at depending on the way you frame  
your answers you could get completely  
different ones

I guess usually will just be slightly  
different answers you can get so these  
are them sort of illustrating the  
challenges that we're facing what  
question are we actually answers asking  
here and why do we get different kinds  
of answers the constructivist point of  
view would say it doesn't make sense to  
argue that we just move our telescopes  
in the basement metaphorically speaking  
in the social sciences because there is  
no basement to go to there is no perfect  
independent point of view and and as  
long as we believe that we're just  
fooling ourselves we need to be trying  
to be compared at becoming aware of the

way that I'm framing my my questions and  
my answers I might get out of it and  
this is in a survey reason resource is  
its qualitative research or this could  
also be any kind of archival research on  
financial stock markets or whatever it  
is I have to be aware of that we've  
created the institutions that we're  
investigating so this is an also leads  
to the different Guba quotes that I've  
highlighted down here and then for  
instance what I've just mentioned here  
that that the was also an inquiry always  
shaped by the interaction of inquiry and  
inquire into that's from the  
constructivist point of view and the  
postpositivist is this sort of agreeing  
with that but being a bit more  
optimistic about how how that there is  
in a big objective out there that we can  
try to get to this isn't the  
methodological implication that I would  
just talked about before so here you  
really don't discover any connections or  
even discover any data you create and  
construct the data set and find answers  
in it so these are the different  
perspectives and how Guba tries to  
Another example  
outline them let me try to use another  
example to make sense of what's what's

going on here any kind of example I could have chosen but I think this one is meaningful because we can intuitively relate to it how should we investigate stress well the assumptions we should rely and the aims that we should have properly dependent needs that kind of question we're interested in okay we're interested in how stress influences job performance let's say this is the overall question but how are we going to measure this I mean we could use smartwatches or any other kind of physiological approach where we want to measure the blood pulse or the heart rate or we want to we could even measure how hormones are being diffused in the body in a stressful situation this here we could argue that there is reasonably clear-cut definitions and measurements that we have we have very reliable instruments that can help us measure these things in a postpositivist this sort of assumption an idea would make sense this is what we try to do we try to build the best possible machines that can help us make the best possible measurements of people's stress at the moment and then use an independent variable in an attempt to explain

people's varying job performance but but  
let's say that we are actually in these  
examples here let's say that we're more

interested with how people perceive

their own stress situation and what

they're trying to do to cope with their

stress and how these coping mechanisms

might influence job performance now we

certainly have to define how they

perceive these job situations and how

they define and then perceive their

coping mechanisms it suddenly becomes

much less clear that we can just say

look let me measure your heart rate this

is the exact same kind of way we

measured across all kinds of human

beings it's only becomes a lot more

heterogeneous if we're going for

people's coping mechanisms and what they

perceive and suddenly we might have to

do a go into some more qualitative

approaches or read their Diaries or

whatever kind of data we could get

access to maybe observe people talk to

people while it happens sort of the

Bernstein approach of putting observers

into a company and trying to track

what's going on so

but we can see that we sort of have the

same overall question job performance

and stress but depending on how we frame

it and how what and maybe also how

optimistic we are in terms of is it

possible to get a completely

standardized measure of what I'm

interested in well if I can and this is

of course what I should do but but what

phenomenon am I dealing with here is it

just the physiological response or is it

my self perception and how I might

respond to different situations and and

then we risk that however questions are

framed my influences so post positivist

what sort of an positivist would push

towards these sort of more standardized

instrument based approaches and then

this is then partially depending on how

optimistic and what kind of question you



want to answer but it's also depending

on exactly the kind of research

questions are you asking okay that was

just another example quite similar in

style or the way that I want to

illustrate this as the breastfeeding

example that I've also uploaded before I

also just want to give you one more

Blueprint metaphor

metaphor to try to make sense of this so

sort of the blueprint metaphor get to

what I mean with that positivism is the

thing that there are facts out there and

and this is what we're trying to

understand postpositivism and and

constructivism are then saying are then

they're having a slightly

different take on this question by a

ricky gervais that we've dealt with a

couple of times what would happen if all

our theory books are burned and can we

recreate the same theories

well postpositivism would say well there

is an underlying blueprint just like a

blueprint for a car there is something

there is a reality underlying how we

make how the world functions how the

business economics world functions that

we could recreate since there is the

same on the line blueprint we should

also in theory get to the same kind of

answers postpositivism would say that

since there's different researchers and

different approaches and different

perspectives it might not be the exact

same answer but there is the same kind

of ideal that we can aspire to

well constructivist would say well first

of all future societies might simply be

different

we might have constructed different kind

of economic environments and different

kinds of ways of investing money and and

in different ways business is

interacting and on top of that we might

theorize completely different about it

there isn't and and this theorization

would also help create the world that we

that we have whatever we sort of now we

have a supply demand mechanism that's

very much linked to the way the

financial and commercial world is set up

but what if we create a different kind

of theories to make sense of human

behavior might be then also create

different kind of societies that would

appear different and would lead to

different kinds of theories so this

blueprint metaphor sort of on a very

abstract level then also illustrates how

postpositivism and constructivism would

would disagree and on a sort of very

experiments as a response to a question

I got because it boasts something useful  
in summarizing the key message in Nelson  
it highlights some the key aspect the  
key challenge of experiments and says  
something more general about it the  
difficulty of doing science so first

What was it

just a quick reminder of what was it

Nelson's 2016 text 3 was about and and  
as a contrast to business economics this

text highlights what classical physics  
and to some degree astronomy is about  
it's about studying for instance atoms

then are just in all respects basically  
identical a hydrogen atom in denmark in  
south america on jupiter anywhere else  
in the universe is just the same it's

absolutely identical we can look at a  
billion atom of them in this room and  
they will be identical and and one could  
sort of make a similar point with

planets even though there are actually  
quite different if we what just want to  
predict how and where a planet will move  
where will it be a month from now we

need the speed and the mass and the size  
and that's it we would we can sort of  
ignore all sorts of other factors so we  
were actually in a situation in physics  
where the challenge of doing science is  
in many ways easier because an atom is

an atom and there is not a lot of stuff  
or and heterogeneity that you have to  
care about and that's the contrast that  
he wants to create to the field of  
business economics we are dealing with  
people and companies just different in  
all sorts of respects of course human  
beings are similar in many ways but  
there was no human being that is as  
identical to another one as two atoms  
are identical to similar atoms identical  
to each other so there's a lot of  
heterogeneity and we cannot say  
something is just the same so whenever  
we sort of want to make a prediction or  
comparison about something  
and we're not comparing identical things  
so we don't have these sharply defined  
boundaries second in contrast to what  
we want to predict our planet moves  
around there are all sorts of things  
that influence friends is how human  
beings or companies behave and what they  
do so it's a number of things that can  
influence numerous they're highly  
variable it's actually difficult to  
separate them from another so we can try  
to define innovation or unemployment or  
similar concepts but there is a genius  
heterogeneity it can be unemployed in  
many different ways and making a really

clear-cut boundary is actually quite difficult and what influences a company is just our human being it's just an immense range of different things for just a relatively large company we could easily make a list of hundreds and hundreds of things that could potentially influence that performance so business economics is just a fundamentally different kind of thing and on top of this we could even have human beings that change behavior over time the subject matter they study often changes over time and add some five billion years ago it's just a hydrogen atom it is basically the same now it just hasn't really changed so and again this is just to sort of reinforce his point that that whenever we talk about supply demand or unemployment or innovation or whatever constructs think we're measuring that we are dealing with there is some kind of construction going on we can't just say look it's right there this is an atom and this is identical to this you can point to one unemployed person that this might be someone who is not searching as acts of E for a job as another unemployed person who might be unemployed for a different reason so so this is the heterogeneity



that the Nelson is talking about and then he's actually making an explicit comparison and link to experiments because the question is how for more genius and similar and predictable is this social world

let's say you want to compare how atoms react to some kind of treatment well again they're identical in one treatment group and a control group but how similar are two classrooms have different teachers and different buildings and consist of different students different cities so it's not as homogeneous so in the elementary school study I guess the advantages was that we had all schools in Denmark at all all the students in of these schools in Denmark so all these small differences in the end even out but it's still just a different setup this is the nulls what he addresses here on page 1699 then it's becoming more popular to do these experiments just as you do in medicine but as he says here for instance if we're talking about experiments in school settings there might be differences between schools or even between classrooms so and almost always students vary in their responses to different treatments so two different

studies are likely to lead to two different results due to this heterogeneity again if we compare two medicine or biology on other fields we could find mice that are just basically identical and live the exact same life this is a challenge that we face this is a challenge that the Nelson is trying to emphasize here on this page in particular were also just in general and Heterogeneity that then leads me to the question that was asked does Nelson oppose to what a good experiment should contain so in other words he says that social science consists of these heterogeneous these different individuals groups and organisations where in an experiment the samples you compare should ideally be as similar as possible and on top of this Nelson says social behavior can change over time so might it make it problematic to be able to do a control study and and and well no he doesn't oppose as such but he just points out challenges to how to how to interpret a randomized control trial and how certain we can be that a result in one setting will also function in another setting and and and we really can believe the both internal and external validity of it so emphasizing that sure if you have really

homogeneous variables like in the Ariely Bionicle experiment where you only had male Howard students and measure something very simple yeah maybe but one should really be transparent and acknowledge that maybe I'm studying something where I struggle with making a clear-cut definition of my variable there might be other things that are influencing that I just don't really I'm not really able to capture so in some sense I think the implication then also is you should try out different variables and different setups rather just saying that a certain cartoon is funny in this or that way as an inverse miling study well maybe we should try out different cartoons with different kinds of people and measure stuff in different ways in order to get to this that robust results we can say look now we found something saying with how meaningful meaningful something is it's maybe we shouldn't just do it Bionicles maybe we should do it with other setups with other samples other people etc so that's really this heterogeneity that he's talking about that we shouldn't think that one simple way of doing it captures everything that we want to know it could in it when we're dealing with

atoms and simple stuff people  
and organisations aren't simple and as also  
said in the beginning that we also  
should be aware of this change in  
behavior so let's just take the  
coronavirus as an example it's going to  
impact how companies and in particular  
large companies are trying to protect  
themselves against this in the future  
maybe ensuring that setups for doing  
work from home how their IT system works  
or how they will deal with customers and  
how close together tables will be in  
restaurants or whatever it is it will  
certainly lead to changes in behavior  
which could mean that whatever we found  
out before is not necessarily true  
afterwards maybe there are some mechanism  
and some behavior of people that  
actually have changed all the time um so  
so this is again just to sort of it's a  
crude example emphasize that just  
because we found out that people behaved  
in this way in 2019 it's not necessarily  
and as a sort of obvious fact true that  
they will behave in the same way in 2020  
we need to do studies in order to figure  
that out this is a challenge that don't  
have in physics the planet Jupiter  
behaves is that one month and one year  
and a billion years ago and an atom

hydrogen atom behaves the way it did  
yesterday we could go 1 million years  
ago and a billion years ago there's no  
change in behavior so that was my  
response to this question about no  
Nelson and experiments and all just in  
general and attempt to try to give a  
brief summary of the key elements